Machine Learning–Based Virtual Sensor for Automotive Engine Combustion Monitoring

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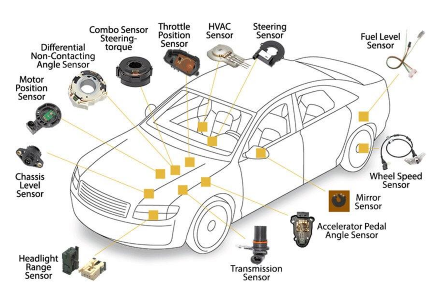
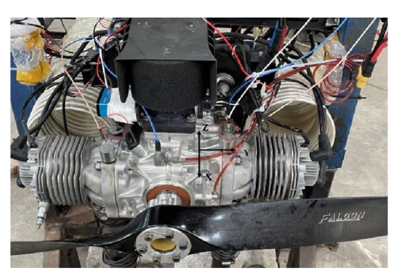
Introduction

- Modern engines require **accurate real-time measurements** for efficiency, emissions control, and regulatory compliance.
- Key variables (MF_IA, NOx, SOC) are: Expensive to measure, Prone to drift, aging, and environmental sensitivity
- **Traditional hardware sensors** increase:
  - Cost + system complexity + maintenance burden
- **Virtual sensing** offers a scalable alternative:
  - Uses existing ECU signals
  - Reduces hardware dependency
  - Improves reliability



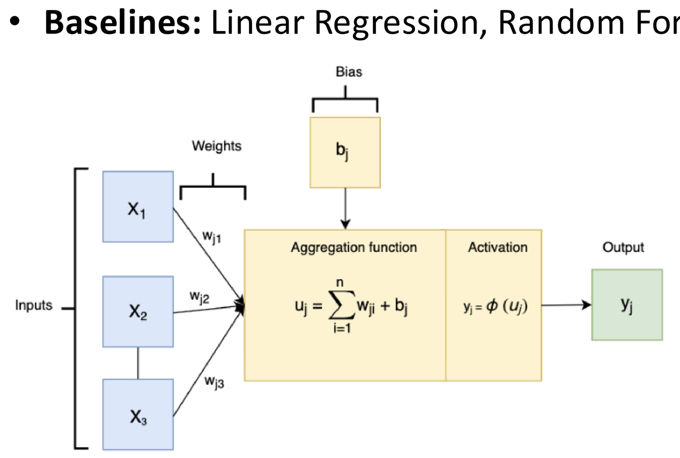
Aim

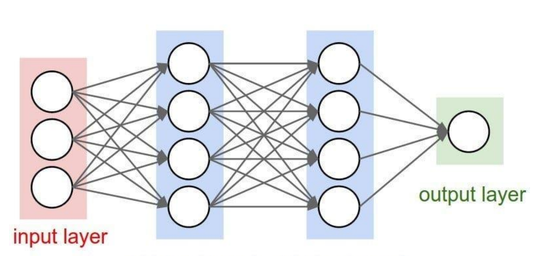
- Develop a **multi-target virtual sensing framework** using a single artificial neural network (ANN).
- Simultaneously estimate:
  - Intake air mass flow (MF_IA)
  - NOx emissions (NOx_EO)
  - Start of combustion (SOC)
- Replace multiple physical sensors with a **unified, software-based model**.



Methodology

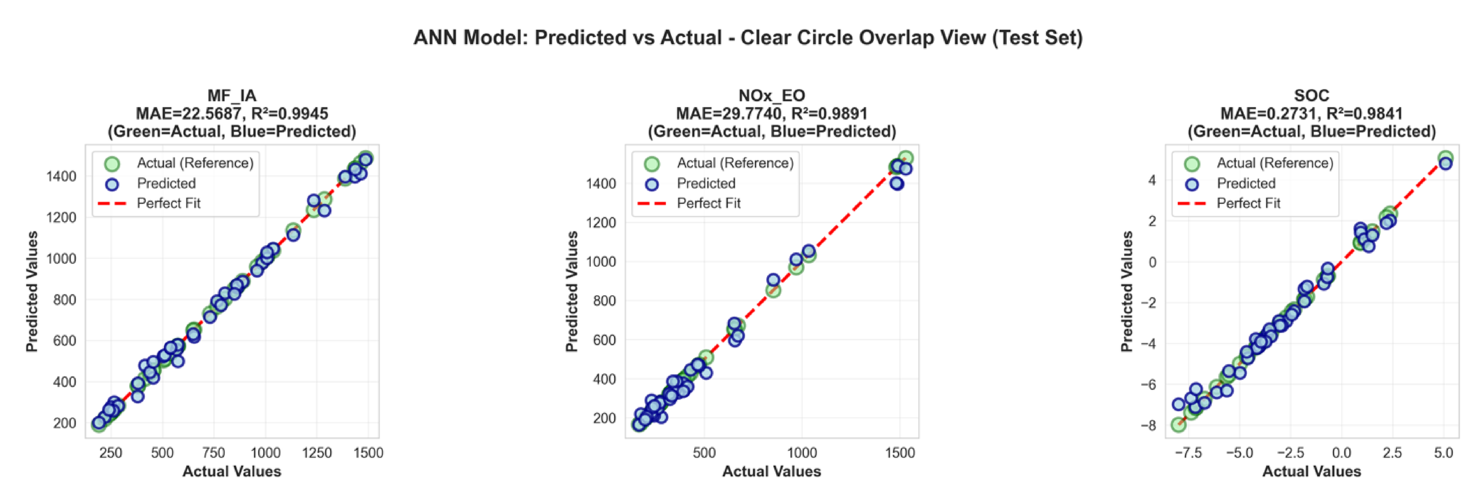
- **Dataset**
  - •12L 6-cylinder diesel engine (real-world data)
  - •**217 operating points** across full load/speed range
  - •Train/test split: **80% / 20% (stratified, seed=42)**
- **Inputs**
  - •**Key inputs (6):**
    - Torque, ambient pressure (P_0), intake temp (T_IM), intake pressure (P_IM), EGR rate, VTG position
  - •**Extended inputs (13):**
    - Add fuel flow, pressures, temperatures, injection parameters
- **Outputs**
  - •MF_IA (airflow)
  - •NOx_EO (emissions)
  - •SOC (combustion timing)
- **Model**
  - •**Multi-output ANN (shared representation)**
  - •Architectures:
    - ANN-6: 6 → 64 → 32 → 16 → 3
    - ANN-13: 13 → 128 → 64 → 32 → 3
  - •ReLU activation (hidden), linear output
- **Evaluation Metrics**
  - •MAE, RMSE, R^2
- **Baselines:** Linear Regression, Random Forest, Support Vector Regression (SVR)





Results

- The multi-output ANN achieves high prediction accuracy across all targets:
  - test **R^2** values of 0.9945 (MF_IA), 0.9891 (NOx_EO), and 0.9841 (SOC)
  - The corresponding **MAE** values (22.6, 29.8 ppm, and 0.27°CA) **meet the required performance thresholds for engine control applications**.
- The predicted vs. actual plots show that all points lie close to the diagonal, indicating accurate predictions across the full operating range with no significant bias.
- Compared to baseline models, the ANN consistently performs best. Linear regression fails to capture nonlinear behavior, while random forest and SVR improve accuracy but remain less reliable, especially for NOx and airflow prediction.
- The close agreement between training and test results indicates good generalization with minimal overfitting. Including additional inputs provides only modest improvements, suggesting that the key inputs already capture the main system dynamics.

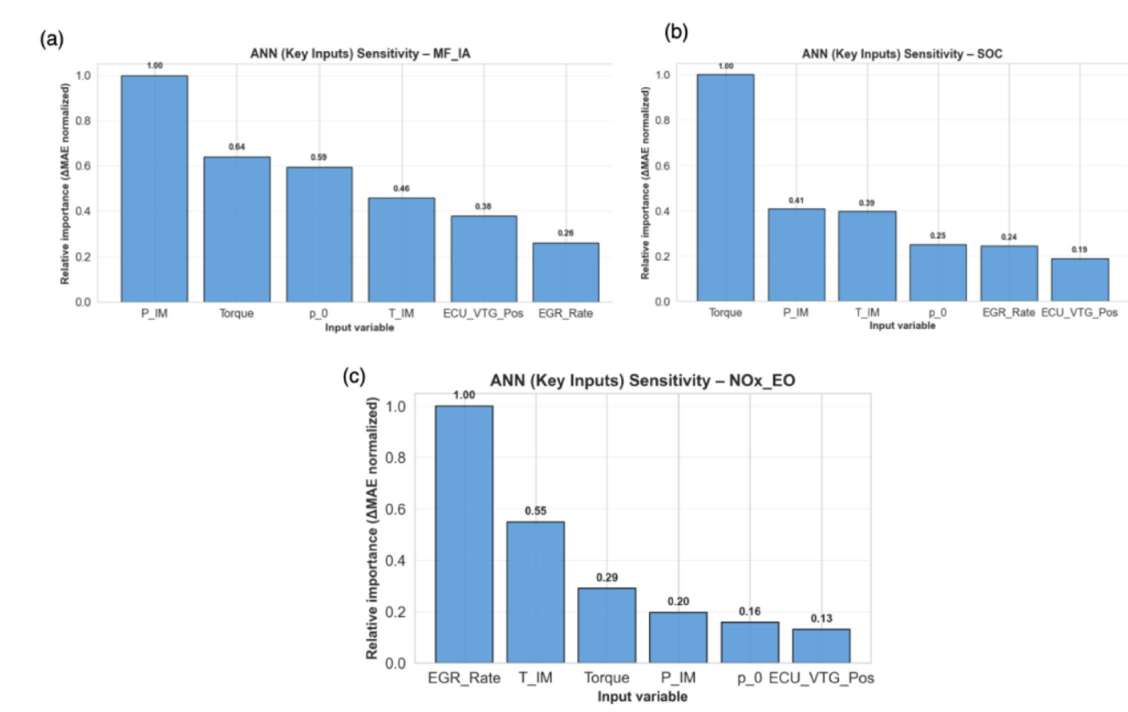


Target	Metric	OLS	Random Forest	SVR (RBF)	ANN (6 inputs)
MF_IA	MAE	49.9	40.4	162.0	22.6
	R^2	0.973	0.981	0.647	0.995
NOx_EO	MAE	89.9	37.5	156.7	29.8
	R^2	0.878	0.974	0.244	0.989
SOC	MAE	1.30	0.40	0.22	0.27
	R^2	0.680	0.958	0.984	0.984

	Airflow (MF_IA)	Nox Formation	SOC Phasing
Dominant Inputs	• Intake pressure P_IM (ΔMAE 181.9) • Torque (ΔMAE 116.6)	• EGR rate (ΔMAE 203.3) • Intake temperature T_IM (ΔMAE 111.5)	• Torque (ΔMAE 1.878) • Intake pressure P_IM (ΔMAE 0.763)
Physical Meaning	• Intake pressure controls cylinder air mass • Engine load regulates air demand	• EGR reduces oxygen concentration • Temperature strongly influences thermal NOx	• Engine load affects ignition delay • Air supply influences combustion timing

Discussion

- Strong **interdependence between outputs**:
  - •Airflow, NOx, and SOC share underlying thermodynamic states
  - •Feature importance is **target-specific**:
  - •EGR → dominant for NOx
  - •Torque → dominant for SOC
  - •P_IM → dominant for airflow
- •ANN advantage:
  - •Learns **shared latent representations**
  - •Handles **heterogeneous nonlinear relationships**
  - •Avoids redundancy of separate models
- •Extended inputs:
  - •Provide **incremental accuracy gains**
  - •Trade-off between performance vs system complexity
- •Model shows:
  - •**Minimal overfitting**
  - •Strong generalization despite small dataset



Conclusion

- A **single multi-output ANN** can effectively replace multiple physical sensors.
- Achieves:
  - •High accuracy (R^2 > 0.98)
  - •Low error within control requirements
  - •Robust performance across operating conditions
- Key contributions:
  - •Unified virtual sensing architecture
  - •Demonstration of shared-representation learning
  - •Practical validation with real engine data
- Implications:
  - •Reduced hardware cost and system complexity
  - •Scalable solution for **software-defined powertrains**
- Future work:
  - •Larger datasets
  - •Real-time deployment
  - •Integration with control systems